

Data Platform Design: 100M Events/Day with Near Real-Time Analytics

Requirements, scale estimation, Kafka ingestion, Flink stream processing, ClickHouse analytics, S3 data lake, Grafana dashboards, fault tolerance, schema evolution, monitoring, security, and scaling to 1B events/day.

Architecture

Client -> API Gateway -> Kafka -> Flink -> ClickHouse -> Dashboard. Target latency: under 60 seconds end-to-end.

Technology Choices

Go ingestion services, Kafka, Flink, ClickHouse, S3, Prometheus, Grafana.